User Interface Prototyping

Reflection…

What are sketches, wireframes, mockups and prototypes?

Why would we use them to design user interfaces?

User interface prototyping

- Sketches, wireframes, mockups and prototypes are all representations of user interfaces used in user interface prototyping
  - We can refer to them as **prototyping methods**

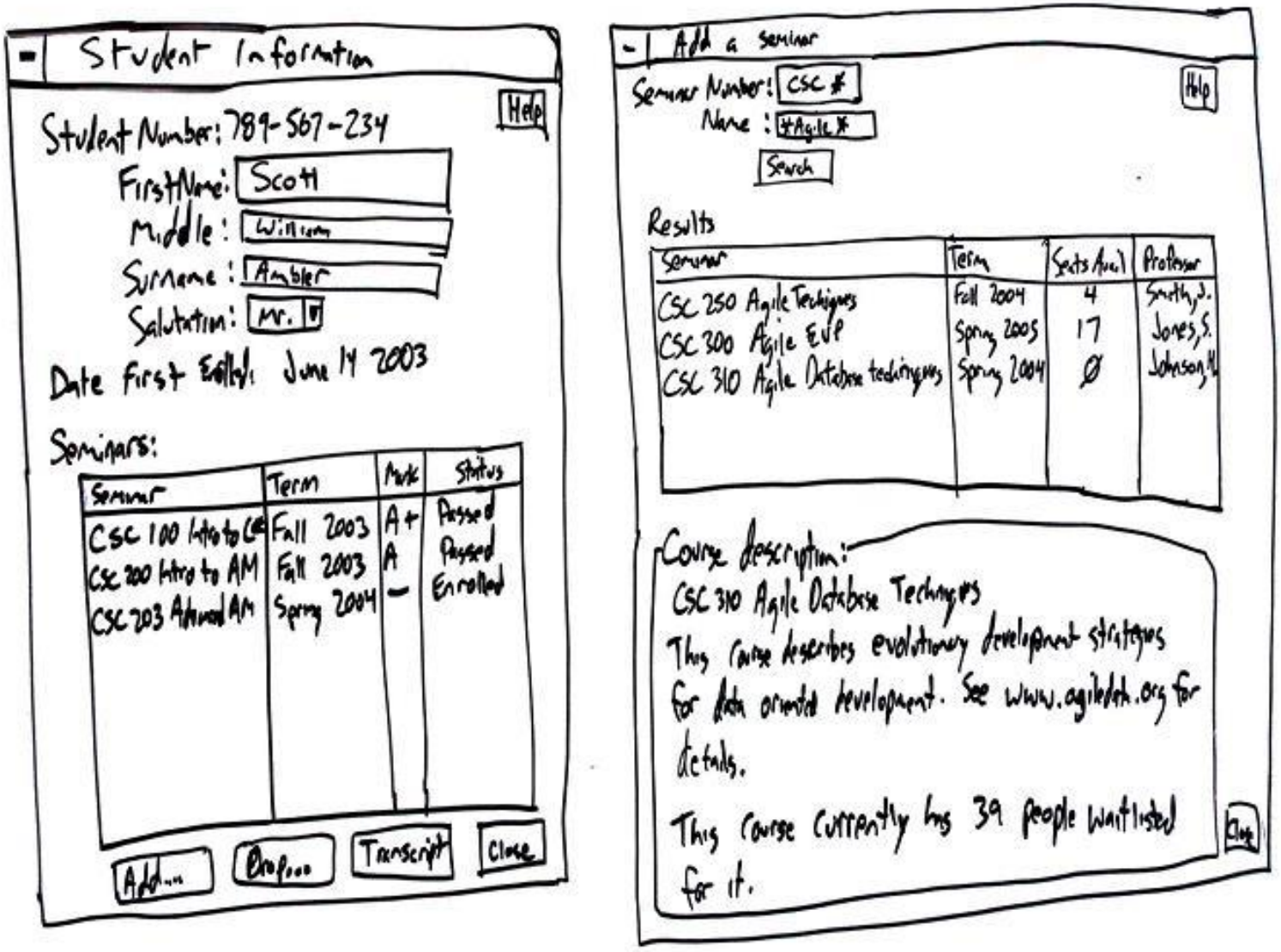
- **User interface prototyping** is an iterative development technique in which interface sketches, wireframes, mockups and prototypes are created
  - Users themselves are typically active in the process of creating and evaluating these design artifacts

What's the difference?

- Unfortunately, in industry, textbooks, and online articles the terms sketches, wireframes, mockups and prototypes often get used interchangeably
  - Are sketches just a form of wireframe?
  - What's the difference between a mockup and prototype?

- We're going to use a more proper granular breakdown, but just be aware of this...

Sketch example



Source: http://agilemodeling.com/images/models/uiSketches.JPG

Sketches

- Sketches are hand-drawn representations of user interface designs
  - Can be done with paper and pencil/pen/markers
  - Or can be done with whiteboards, blackboards, etc.
  - Or as in the saying, the "back of a napkin"

- Sketches have ***virtually zero barrier to entry***
  - Even a non-technical customer/user can sketch
  - Sketches can be created easily ***while with*** the customer and as a group

- Cost in time to create sketches is very low
  - Sketches need to represent idea effectively, they don't need to look good and polished (straight lines, colours, etc.)

Sketches

- Sketches are very low fidelity
  - Prototyping methods often ranked from low fidelity to high fidelity
  - Lower fidelity means low level of detail compared to working product... less time consuming to produce, but less valuable feedback
  - High fidelity means level of detail approaches or matches the working product... more time consuming to produce, but more valuable feedback

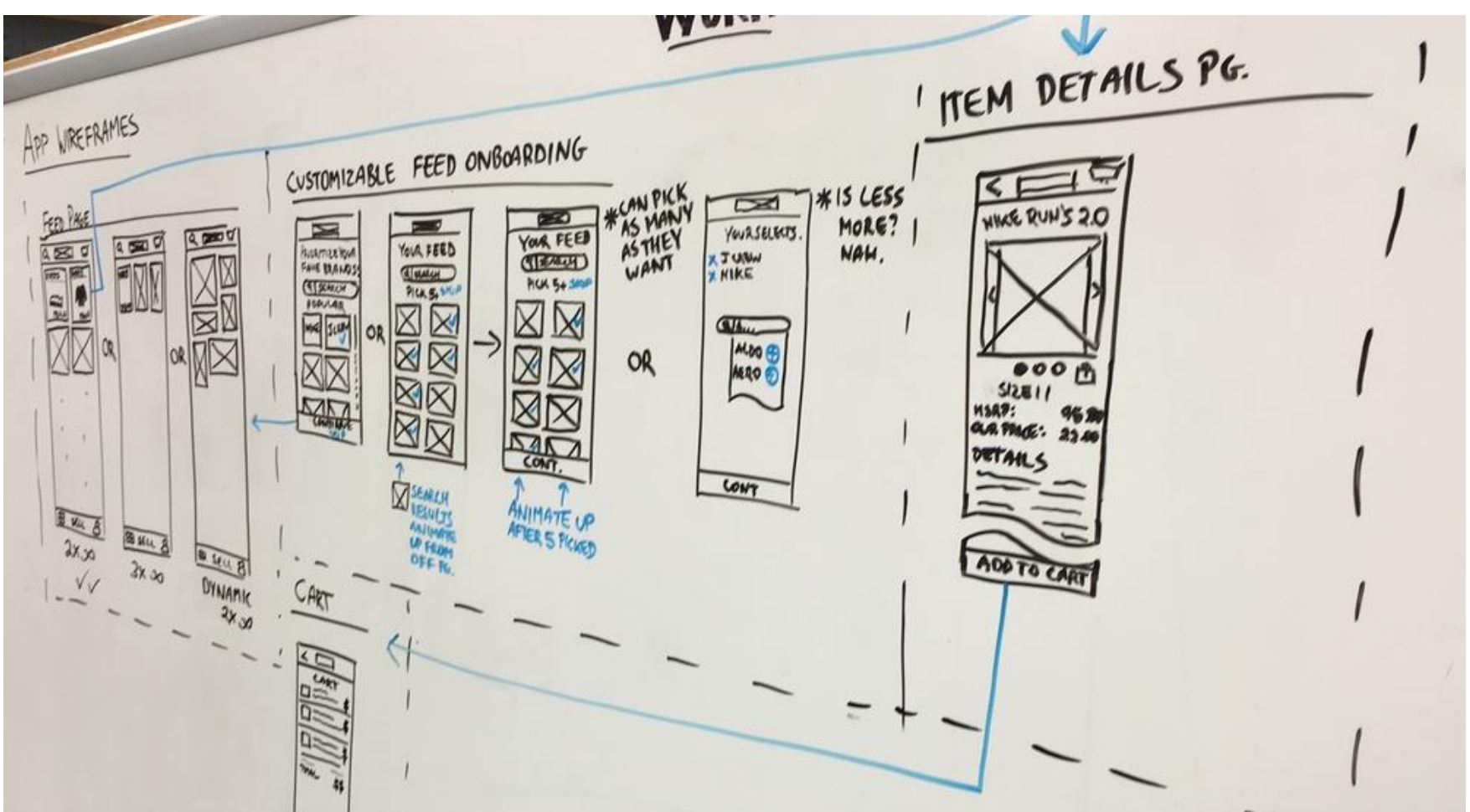
Sketches

- We actually can model interactivity with sketches and other low-fidelity prototyping methods

- CTO of a successful health tech startup once had me "tap" on paper sketches at coffee shop, he then put down the next screen... worked surprisingly well!

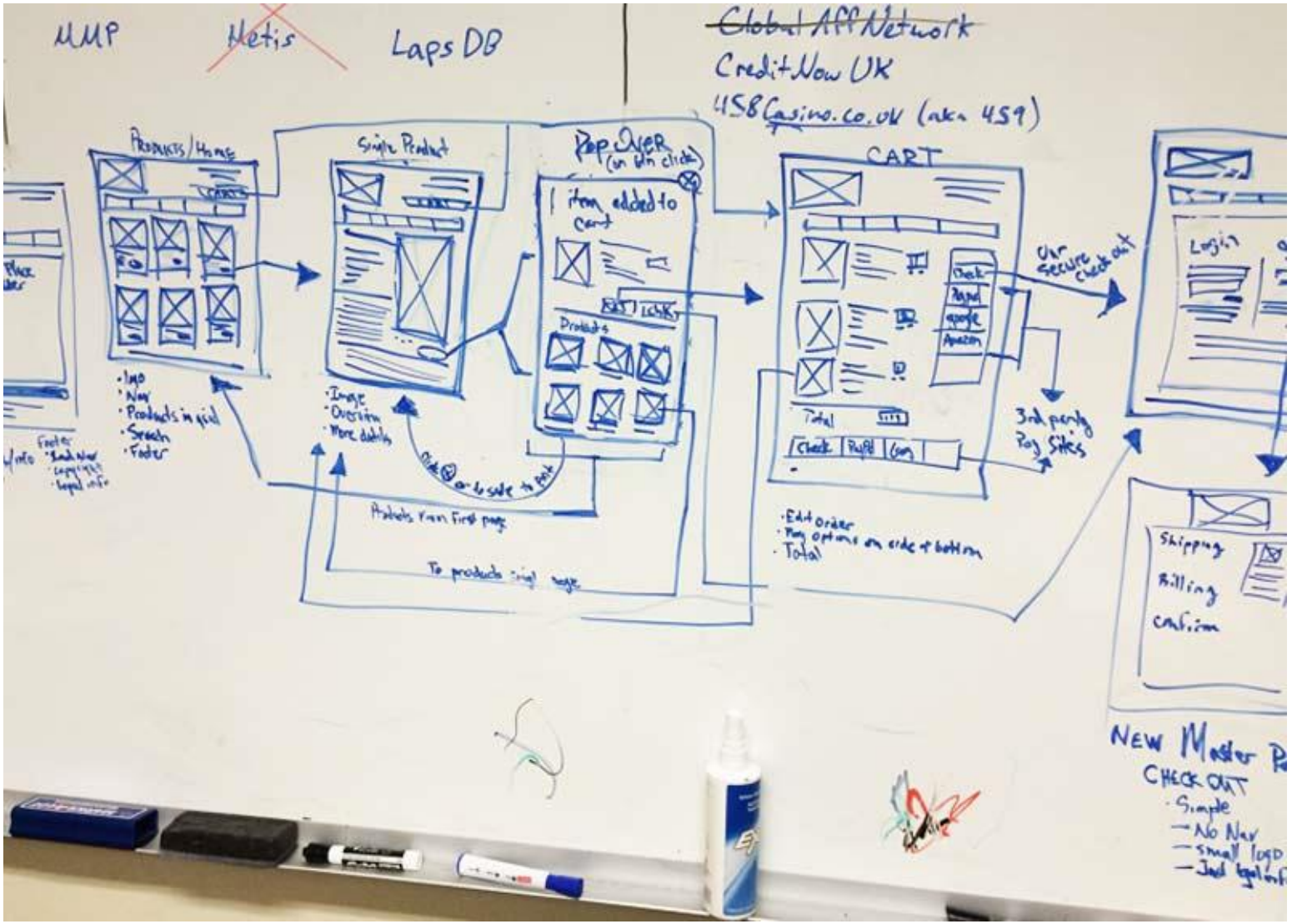
- When using a whiteboard, we can draw arrows from one sketch to the next, change relationships as need be

Whiteboard sketch example



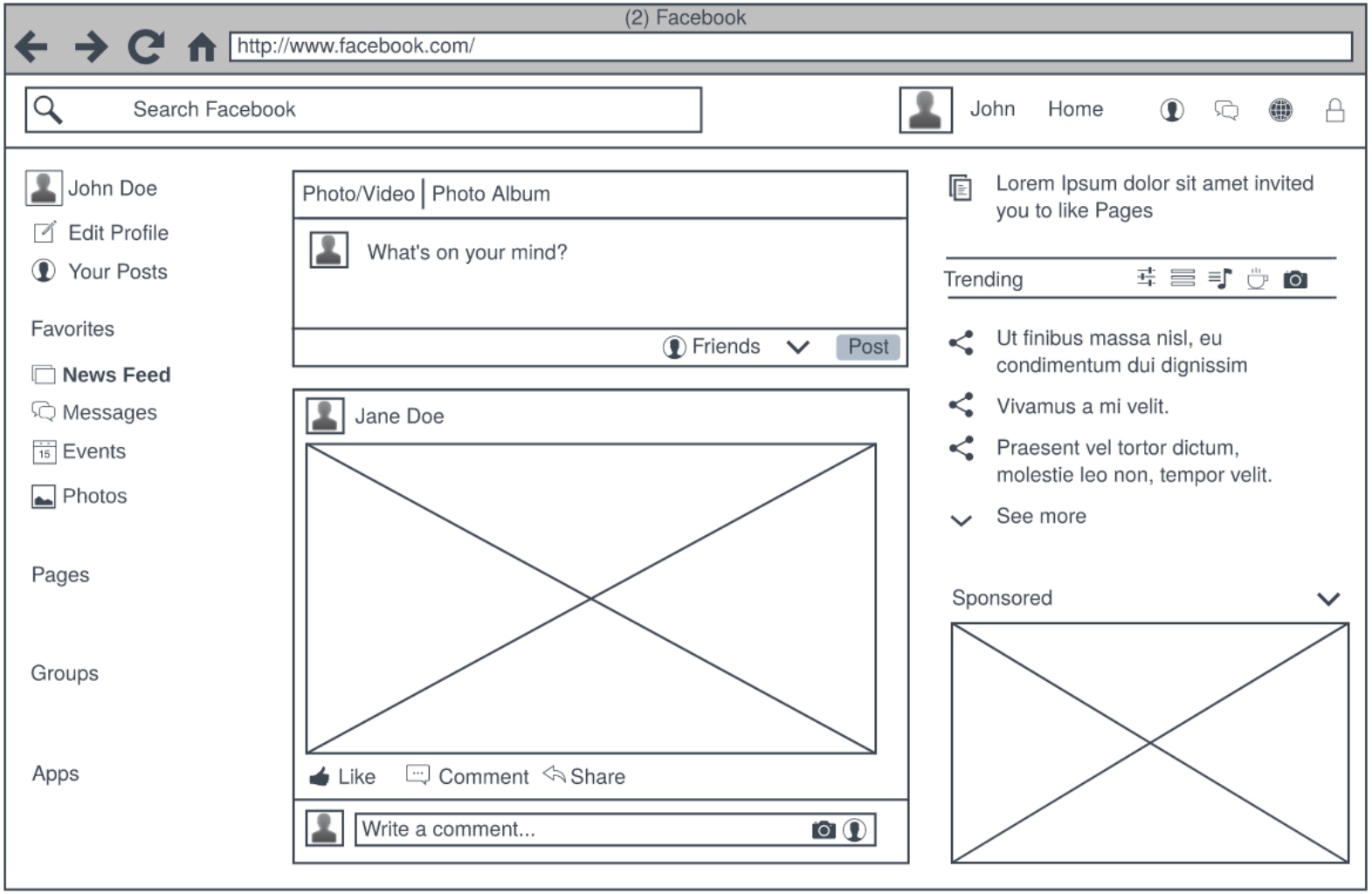
Source: https://images.squarespace-cdn.com/content/v1/55a90ce7e4b074b43a47cdc1/1456268974713-RVAK4BB7ZH70MENXW5NB/ke17ZwdGBTodDI8pDm48kMUWSGNGjkQEncI8WER1P0d7gQa3H78H3Y0txjaiv_0fDoOvxcdMmMKkDsyUqMSsMWxHk725yiiHCCLfrh8O1z4YTzHvnKhyp6Da-NYroOW3ZGjoBKy3azqku80C789l0r9YoV8ytu8SWDj21Bt3yU9PAZ2yiMLVPwT7WG3OaePMdH_2KkHTy5mFnfAgJe0TNw/image-asset.jpeg

Whiteboard sketch example



Source: https://www.np-creative.com/wp-content/uploads/2014/05/nick-pantier-wireframe2.jpg

Wireframe example



Source: https://d2slcw3kip6qmk.cloudfront.net/marketing/pages/chart/facebook-wireframe.svg

Wireframes

- Wireframes accurately represent interface page layout and organization of content
  - But lack color, typography, images and other graphics
    - May have some basic color, placeholder graphics, but not the final version or intended to be
  - Wireframes are analogous to blueprints
- Can be done by hand, but expectation of precision is higher than sketches... software more often used
- Wireframe layouts can be connected to one another to represent interactivity (but not a requirement)

Wireframes

- Increased precision over sketches and use of software makes wireframes less accessible than sketches

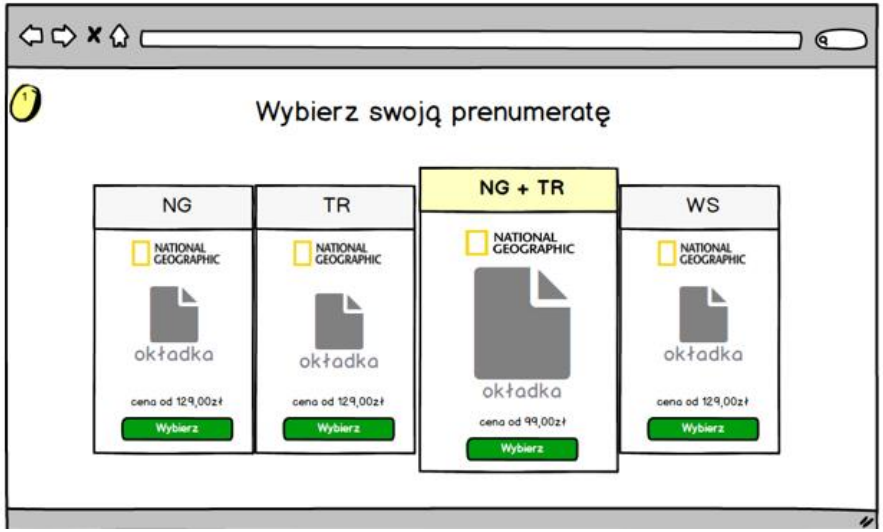
- Wireframes more time consuming to create than sketches, but relatively higher fidelity
  - Working interface should very closely resemble wireframe, certain details aside

- Wireframes provide more concrete representations of designs made as sketches

Wireframe to mockup

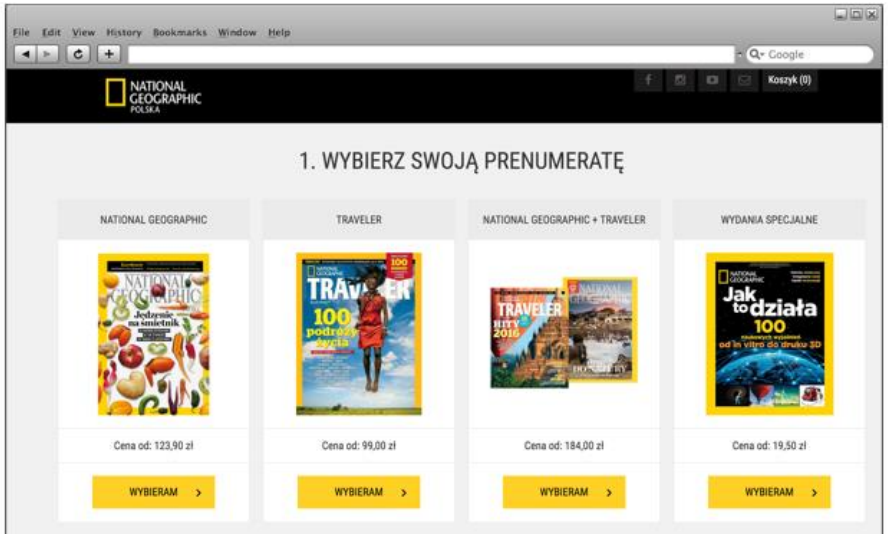
WIREFRAME

Structure + Functions + Content



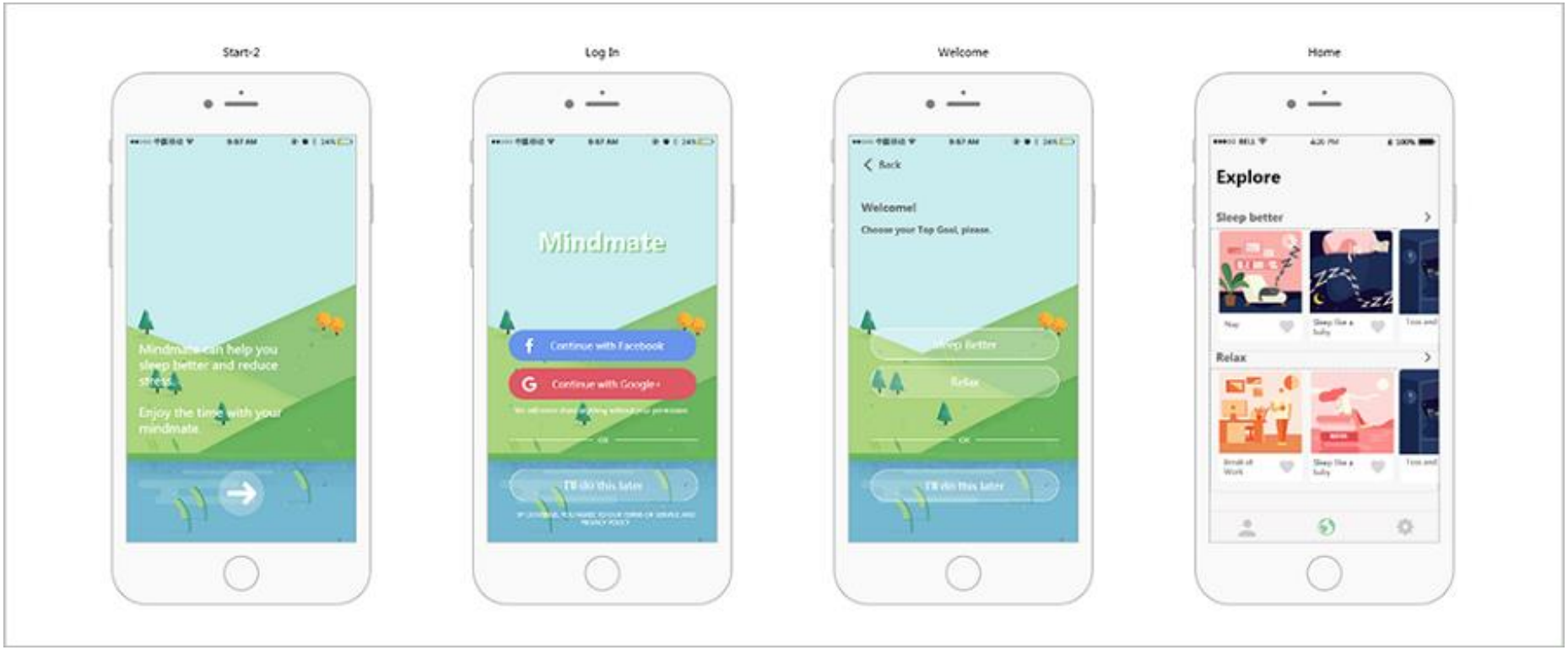
MOCKUP

Style + Colours + Right Content



Source: https://brainhub.eu/blog/wp-content/uploads/2016/04/difference-wireframe-between-mockup.png

Mockup example



Source: https://miro.medium.com/max/1804/0*rZBE0Dj38um6Owdi.png

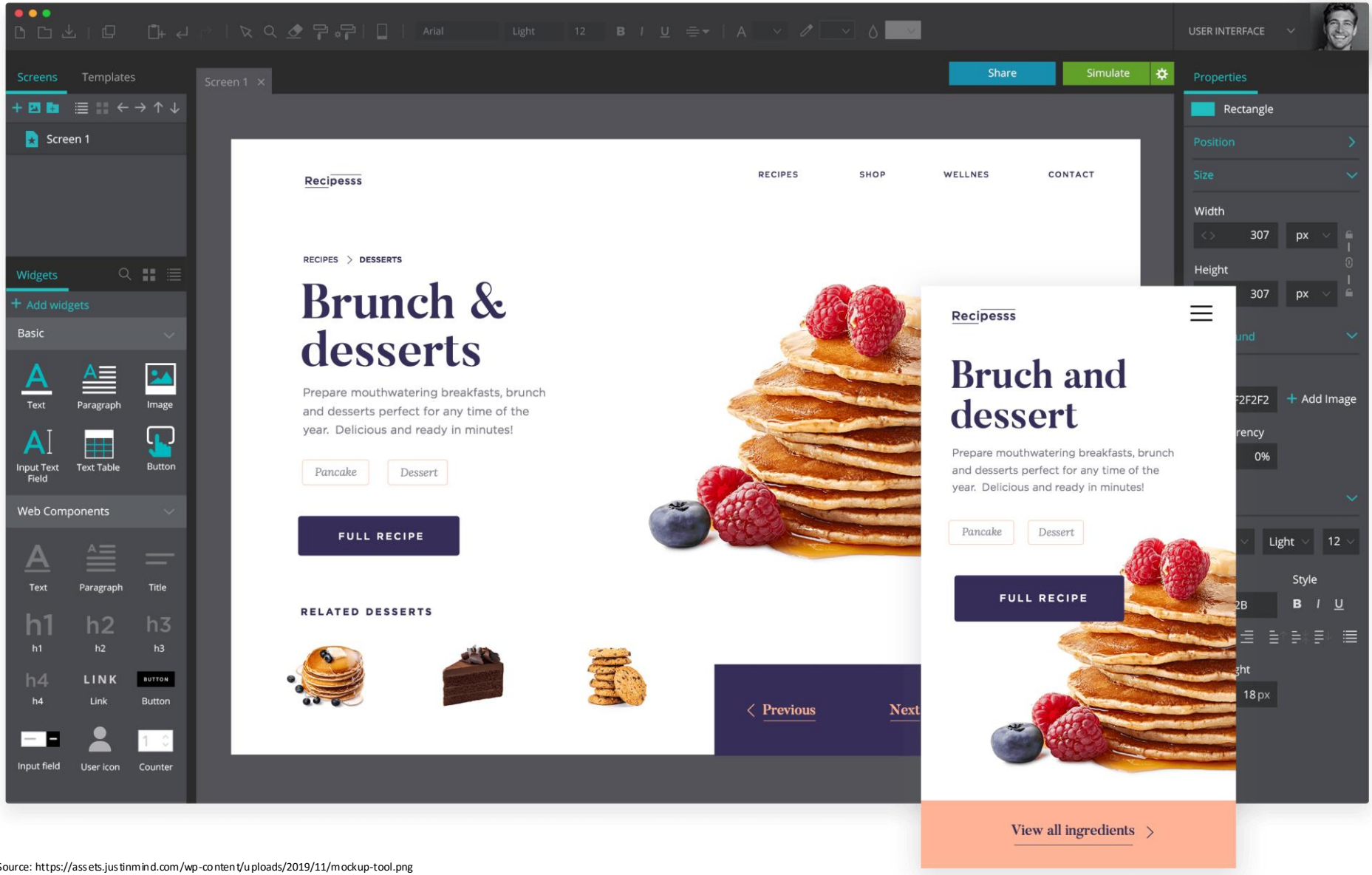
Mockups

- Mockups are static but accurate representations of the application interface
  - Color, typography, and graphics should be included, even if they are samples, e.g. profile images
  - Mockups are not interactive, cannot click and perform actions
  - Equivalent to screen captures of the final interface

Mockups

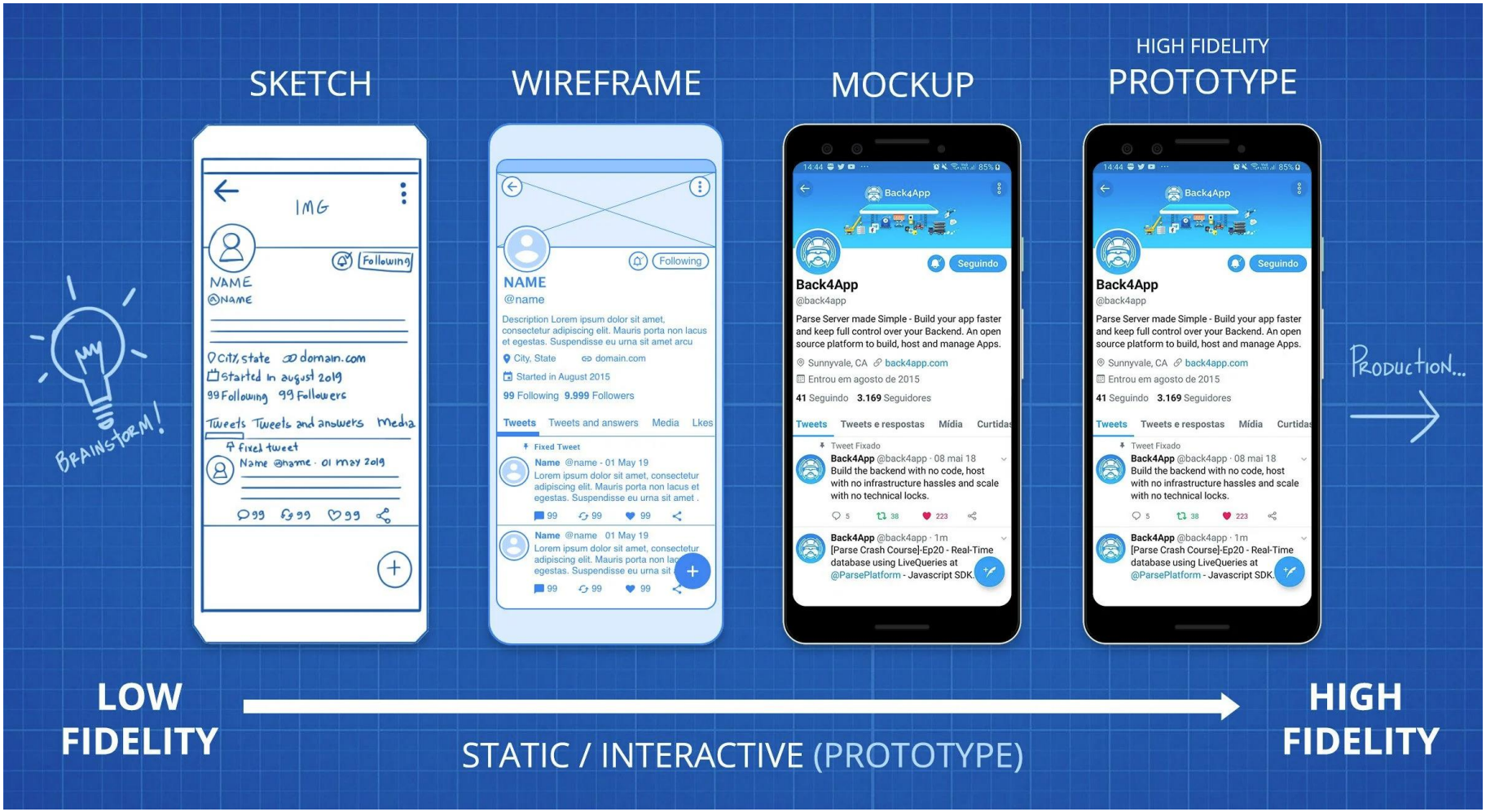
- Mockups are usually created with purpose-built mockup tools or graphics editors like PhotoShop or GIMP
  - You can't sketch a mockup
  - Technically, you can create a mockup with HTML/CSS but usually this is left for prototype...
- Mockups are even more time consuming than wireframes, but very high fidelity

Mockup creator tool example



Source: https://assets.justinmind.com/wp-content/uploads/2019/11/mockup-tool.png

Prototype


SKETCH
WIREFRAME
MOCKUP
HIGH FIDELITY
PROTOTYPE
BRAINSTORM!
PRODUCTION...
LOW FIDELITY
STATIC / INTERACTIVE (PROTOTYPE)
HIGH FIDELITY

Prototypes

- Prototypes are an accurate ***and*** interactive representation of the application interface
  - We can perform actions in a prototype and the result should be the same in the full working application (e.g. transition from one screen to next)

- Difference between prototype and completed application?
  - Not connected to a back-end and all of the related logic (e.g. login, user authentication, etc.)
  - May use hardcoded data

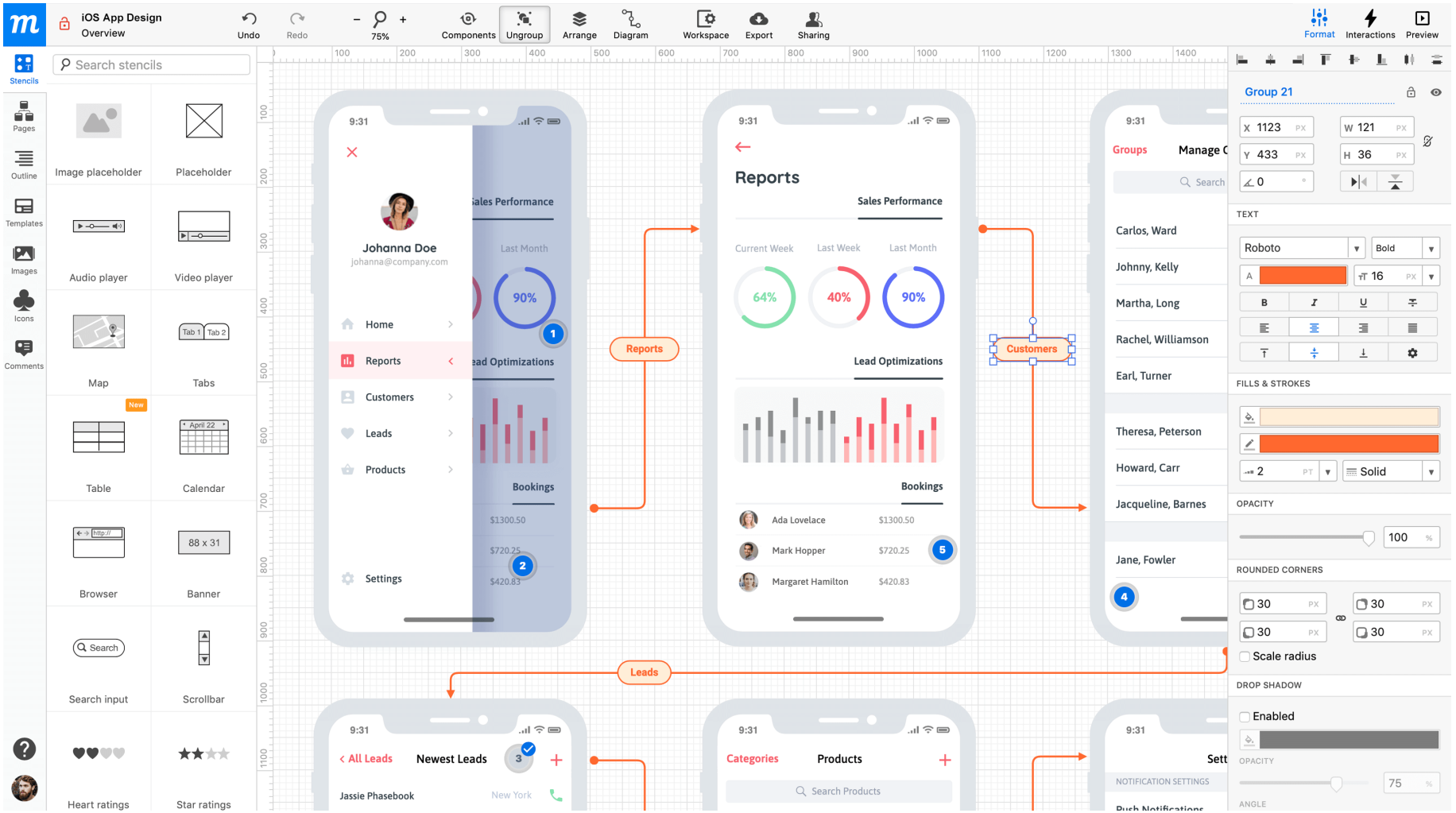
Prototypes

- Prototypes are the most high fidelity, most time intensive to create
- Prototypes can be created with the same tools as the completed application
  - e.g. HTML, CSS, JavaScript, etc.
  - Saves development time when final step is to connect to back-end
- Or prototypes can be completed with purpose-built tools that allow for simulated interactivity

Prototypes

- Idea with prototype is that in terms of UI it is fully functional, which has several benefits:
  - We can verify the UI with the customer/user
  - We should be able to build user help using a prototype while the rest of the system is in development
  - We should also be able to use prototypes for sales and marketing purposes

Prototype creator tool example

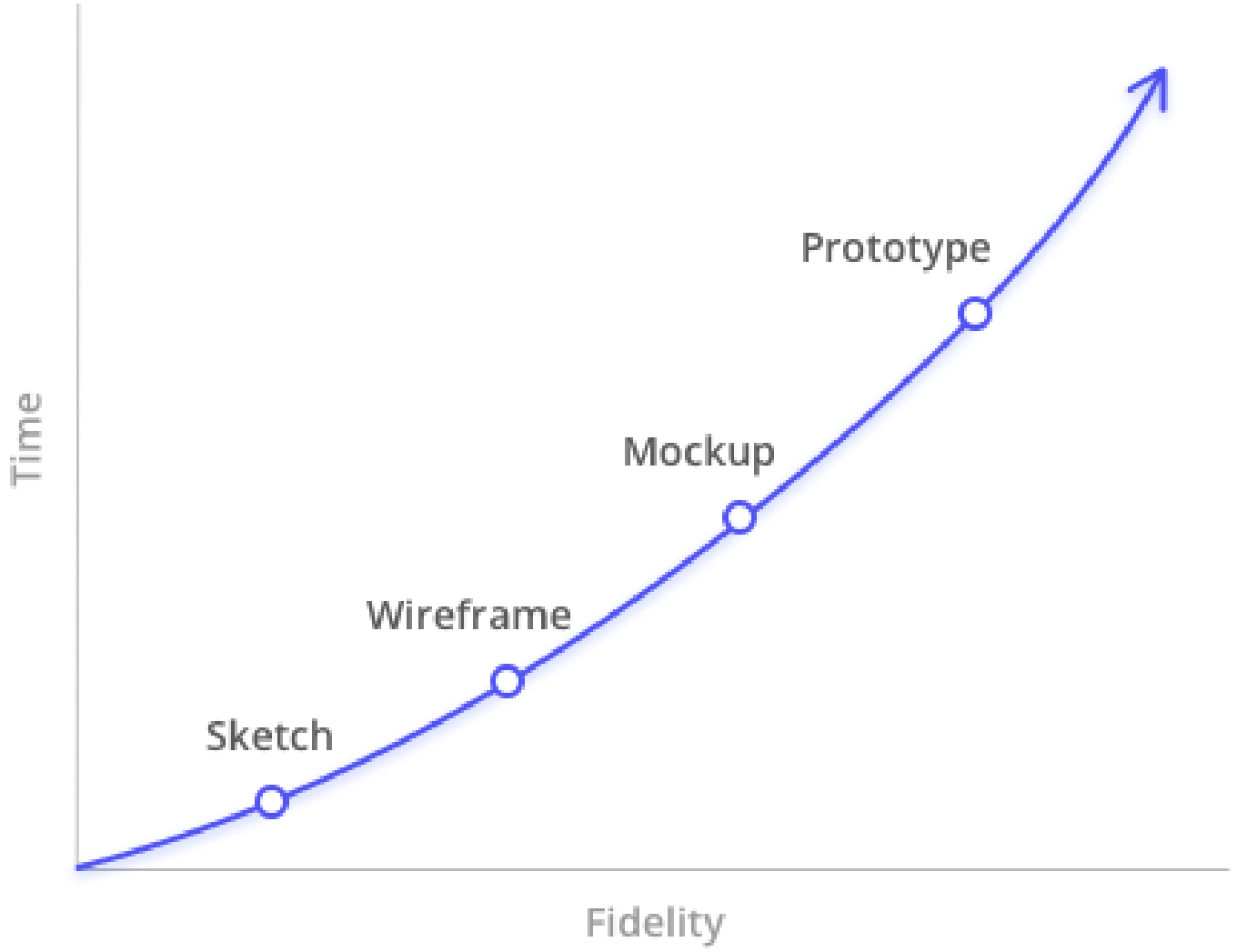


Source: https://landing.moqups.com/img/product-shots/moqups-overview.png

Terminology reminder

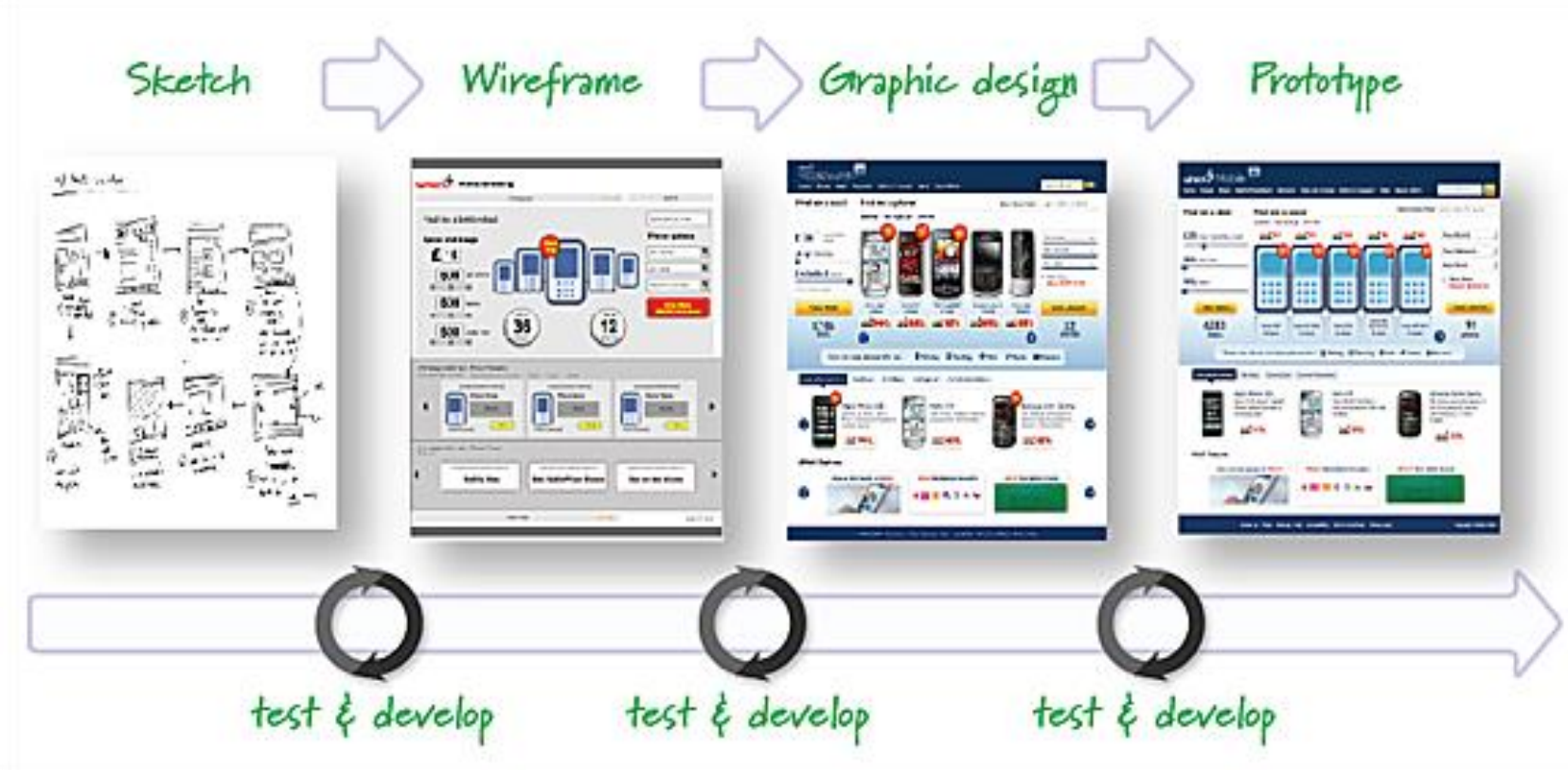
- A reminder that terminology is all over the place in practice
- Low fidelity vs high fidelity is often the distinction that is made
  - Low fidelity vs high fidelity wireframe/prototype/etc
  - Really the most important distinction anyways!
- Sometimes "interactive prototype" or "UX prototype" is used to describe working prototypes

Key relationship: cost vs fidelity



Source: https://uploads-ssl.webflow.com/5e21025d55fb4235889a1e25/5e223bfb72595d77d653ae0a_5ce1d9528eff8b4dfb09c8c1_time-fidelity-wireframe-mockup-prototype.png

Different purposes at different stages of design process…



Source: https://magora-systems.com/uploads/pages/728/wPLJwd_dWZuXt7LvbFGrPVHerS1BHWEY.png

Design process

- Precise design processes vary from organization to organization
  - Sketch -> Wireframe -> Mockup -> Prototype is not "the process itself", but they tend to be used in sequence in a process (requirements gathering -> design -> …)

- In general though…
  - Sketches for establishing requirements and rough UI with client, while with the client
  - Wireframes a firmer "blueprint" based on sketches
  - Mockups give developers something precise to create
  - Prototypes created by developers, final check and working out of issues before full implementation

Tools

- Sketches
  - Pen, paper, whiteboard, etc.
- Wireframes
  - https://wireframe.cc/ [free]
  - http://draw.io/ [free]
- Mockups
  - PhotoShop (https://www.photoshop.com/en) [expensive]
  - GIMP (https://www.gimp.org/) [free]
  - Tools for prototyping double as mockup tools
  - HTML, CSS (possibly libraries)
- Prototypes
  - Tools like Figma, Google Stitch, Moqups
  - HTML, CSS, JavaScript (possibly libraries)

Modern tools

- Figma has become the most popular prototype creation tool by far, we could say it's now the industry standard tool
  - 13 million users as of March 2025
    - https://sqmagazine.co.uk/figma-statistics/
- As of 2026 new AI-based tools for prototype creation are surging in popularity, for example:
  - Figma Make
  - Google Stitch

Make your ideas real with Figma Make

What do you want to make?

Onboarding flow | Data dashboard | Gradient gallery

Make your ideas real with AI. Start with a design and prompt your way to a functional prototype, fast.

Discover Cats

PurrSocial

Search for pawsome cats...

All Paws | Ginger | Calico

Trending Meows See All

#GingerHero
Meet Barnaby: The Sun King

Top Zoomies of the Week

Featured Paw

Discover | Following | Profile

Following Feed (Interactive)

PurrSocial

Me | Mittens | Luna | Oliver

Simba
2 hours ago

#Zoomies

1,248 56

Simba Found the red dot today. It didn't stand a chance! 🐱#CatLife #Hunter

Luna
5 hours ago

#NapTime

892 12

Luna The sunbeam found me today. Best 4 hours of my life. 💤

Discover | Following | Profile

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Cat Profile (Interactive)

PurrSocial

Sir Pounce-a-lot

#NapExpert | #Zoomies | + Follow

12.4k FOLLOWERS | 342 POSTS | 89 FOLLOWING

About Me

Professional box tester and midnight marathon runner. I specialize in knocking pens off desks and demanding treats at 4 AM. Living my best 9 lives one sunbeam at a time. 🐾

Recent Paws | Videos | Tagged

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PurrSocial

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Discover | Following | Profile

What would you like to change or create?

3 Flash

Modern tools

- These tools are challenging the relationship between cost vs. fidelity, and accessibility, etc.

- Even without considering AI tools, Figma Design already allows users to create prototypes with a largely drag and drop interface
  - This lowers the cost of creating high fidelity prototypes compared to before when a prototype application made with tools like HTML/CSS/JS would have to be developed, not as low as low fidelity prototypes, but getting closer to making wireframes

Modern tools

- AI is lowering this cost even further and making the tools more accessible too
  - People only need to be able to write prompts in order to create prototypes, modify them, etc.
  - The prototypes are then created in high fidelity in seconds to minutes, and can be refined with further prompts

- So instead of using sketches when coming up with ideas together with a client, should be now use these AI-based tools instead?

Modern tools

- These AI tools lower the cost of "building", but will that cause teams to overlook the more difficult design decisions that need to be made?
  - It's so easy to make a nice prototype that we may not stop to think about the quality of the design
  - We get something "good enough" fast, but had we used a slower and more deliberate process we might have come up with something even better
- These tools are so new that the rules are being rewritten right now, these are the problems folks in industry are looking at themselves
  - It's 100% OK to not know the answers yet, in situations like this it’s OK to just learn and experiment ourselves